

120 Years of Olympic History: Participation and Medal Trends

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1 Executive Summary

This report looks at 120 years of Olympic history – from the 1896 Athens Games to the 2016 Rio Games - to analyze participation and medal-winning trends. The analysis shows a major rise in the number of women athletes over time, and a continued dominance of wealthy countries in the medal tallies. Australia has a high Olympic performance relative to its population. The

findings highlight the evolving inclusivity of the Olympics and the impact of national resources on sporting success.

2 Introduction

The Olympic Games is the most popular international sports competition that gathers the best sportsmen from all the world every four years. Since their modern revival in Athens in 1896, the Games have expanded enormously in scale, diversity and global importance. Studying historical trends in Olympic participation and performance can shed light on wider trends in gender equality, national development and sporting investment. This report looks at athlete-level data over 120 years of Olympic history to bring out key trends in participation and medal outcomes.

The dataset, from Kaggle (Griffin 2018), contains over 271,000 observations of athletes, sports, events, and medal results from Summer and Winter Games. The analysis is restricted to the Summer Olympics to ensure comparability of similar events.

By analysing both visual and statistical trends, this report aims to answer key questions about how have participation and medal-winning patterns changed across Olympic Games from 1896 to 2016, with a focus on gender representation and top-performing nations

3 Research Questions

The analysis is guided by three key questions:

- How has overall athlete participation in the Summer Olympic Games changed over time?
- Has the proportion of Olympic medals won by female athletes increased over time?
- Has the distribution of Olympic medals among nations become more or less concentrated over time?

4 Data source:

The dataset is sourced from Kaggle and acknowledged as The Olympic data on www.sports-reference.com, which contains information on athletes, their participation in Olympic Games, and the medals they won from 1896 to 2016. The dataset includes variables:

- ID: Unique number for each athlete
- Name: Athlete's name

- Sex: M or F
- Age: Integer
- Height: In centimeters
- Weight: In kilograms
- Team: Team name
- NOC: National Olympic Committee 3-letter code
- Games: Year and season
- Year: Integer
- City: Host city
- Sport: Sport
- Event: Event
- Medal: Gold, Silver, Bronze, or NA

5 Methodology

5.1 Visualisation:

The analysis begins with visual exploration of the data to identify key trends in participation and medal outcomes. Line plots and table are used to describe the trend and highlight the top-performing countries in terms of total medals won.

5.2 Statistical Analysis:

The following statistical models are fitted to answer the research questions:

1. Overall participation patterns over time: To determine whether Olympic participation has increased over time, a linear regression model is fitted: $Participation_t \sim \beta_0 + \beta_1 * Year + \epsilon_t$ where $Participation_t$ is the number of athletes participating in each Year. A significant positive coefficient for Year (β_1) would indicate an increase in participation over time.

2. Female Medal Success over time: To investigate whether female athletes have become more successful over time, the proportion of medals won by women is analysed using a binomial logistic regression model: $\log\left(\frac{p_t}{1-p_t}\right) \sim \beta_0 + \beta_1 * \text{Year} + \epsilon_t$ where p_t is the proportion of medals won by female athletes in each Year. The model is fitted using the yearly counts of medals won by female and male athletes. A significantly positive estimate of β_1 would indicate an increasing trend in female medal success over time.
3. Medal Concentration Among Nations: To evaluate whether medal concentration among nations has changed over time, the Herfindahl-Hirschman Index (HHI) is calculated for each Year based on the distribution of medals among countries. A linear regression model is then fitted: $HHI_t \sim \beta_0 + \beta_1 * \text{Year} + \epsilon_t$ where HHI_t is the Herfindahl-Hirschman Index for each Year. A significant negative coefficient for Year (β_1) would suggest a decrease in medal concentration among nations over time, indicating a more even distribution of medals across countries.

6 Results

6.1 Visualization

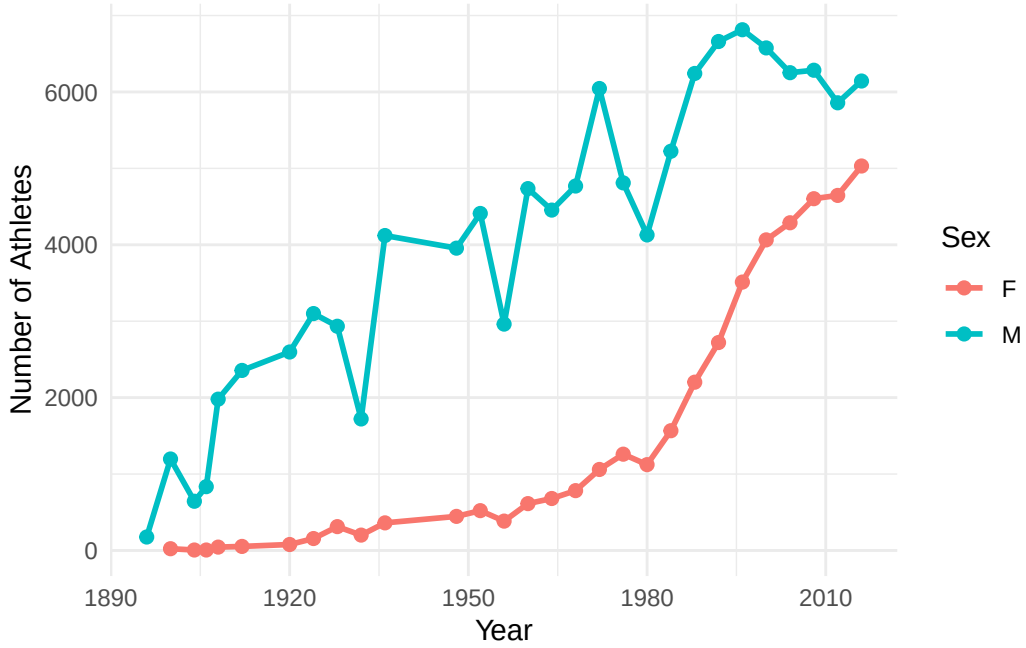


Figure 1: Athlete Participation by Gender in the Summer Olympics (1896–2016)

Table 1: Top 5 Countries by Total Medals (Summer Olympics)

Country	Total Medals
United States	4686
Soviet Union	2061
Germany	1687
Great Britain	1598
France	1408

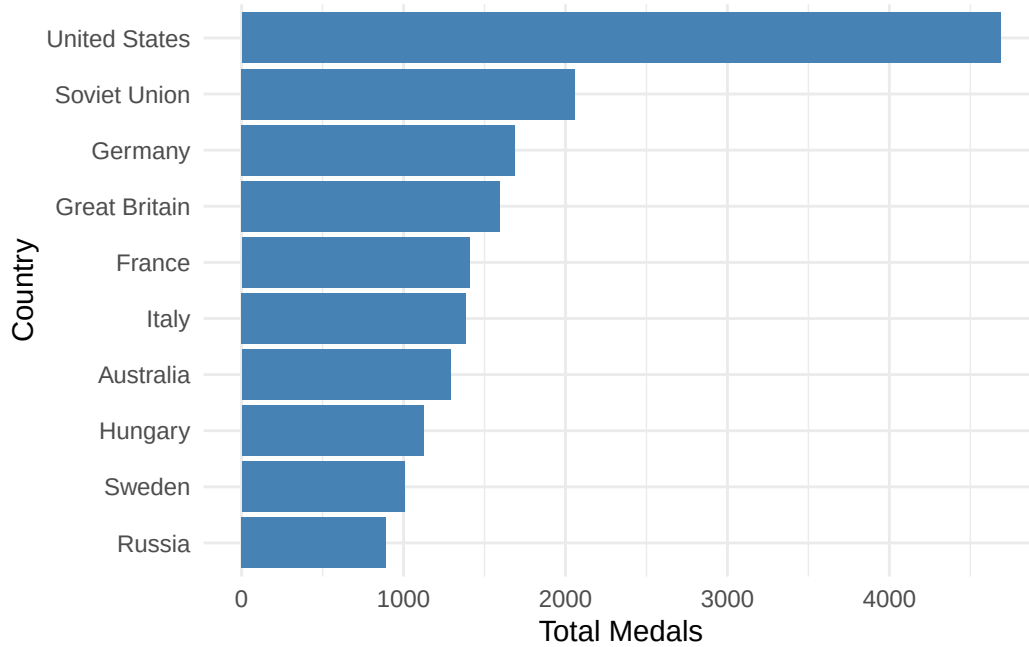


Figure 2: Top 10 Countries by Total Medal Count in the Summer Olympics (1896–2016)

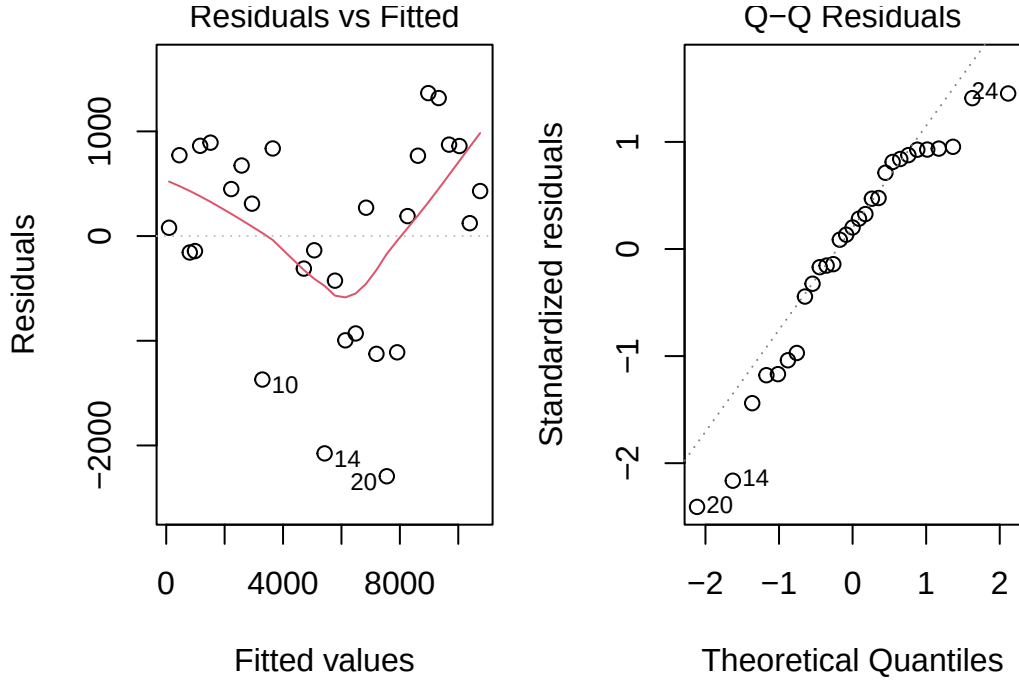
Figure 2 shows the ten countries with the highest total number of medals won through all Summer Olympic Games. Leading is the United States, then comes the Soviet Union and Germany. Australia can be found among the ten most successful countries. There is a noticeable domination of rich countries throughout history, which proves the correlation between national resources and Olympic success.

6.2 Statistical Analysis

1. Overall participation patterns over time:

Table 2

Term	Coefficient	p-value	R ²
Intercept	96.6205	0.7825	0.924784415470121
Year	88.7696	0.0000	



(a) Residuals vs Fitted

Figure 3: Residual plots for participation model

According to Table 2, the coefficient for Year is positive and statistically significant ($p < 0.001$), indicating that Olympic participation has increased over time, around 88 more athletes per year on average. The R-squared value (0.924) also supports that Year explains a large proportion of the variation in athlete participation across Olympic Games.

2. Female Medal Success over time:

Table 3

Term	Coefficient	p-value	Pseudo R ²	Null Deviance	Residual Deviance
Intercept	-3.4954	0	0.964539393668614	5109.44371104006	181.183972009567
Year	0.0302	0			

Table 3

Term	Coefficient	p-value	Pseudo R ²	Null Deviance	Residual Deviance
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Table 3 shows the positive coefficient indicating that the proportion of medals won by female athletes has slightly increased throughout Olympic history, around 3% per year on average. The p-value for Year is statistically significant ($p < 0.001$) and the Psuedo R-squared value (0.964) highly supports that Year explains a large proportion of the variation.

3. Medal Concentration Among Nations:

Table 4

Term	Coefficient	p-value	R ²
Intercept	0.2142	0.0000	0.274710131339156
Year	-0.0016	0.0035	

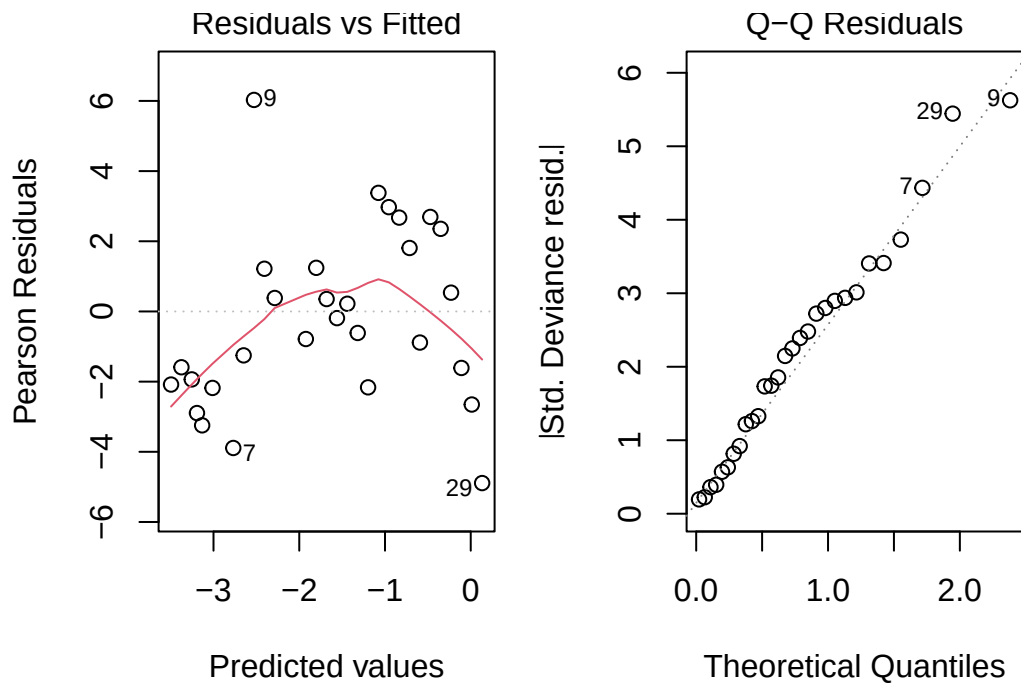
Table 4 indicates a stastically significant decline in medal concentration over time, with a negative coefficient for Year, around 0.16% decrease in HHI per year on average ($p\text{-value} < 0.005$). The R-squared value (0.274) suggests that Year explains a moderate proportion of the variation in HHI across Olympic Games.

Figure 3 and Figure 4 indicate that the residuals are not reasonably random, which suggests that the linear and logistic regression models may not fully capture the relationship. However, Figure 5 look better, suggest that the linear regression model for HHI works reasonably well.

7 Conclusion and Discussion

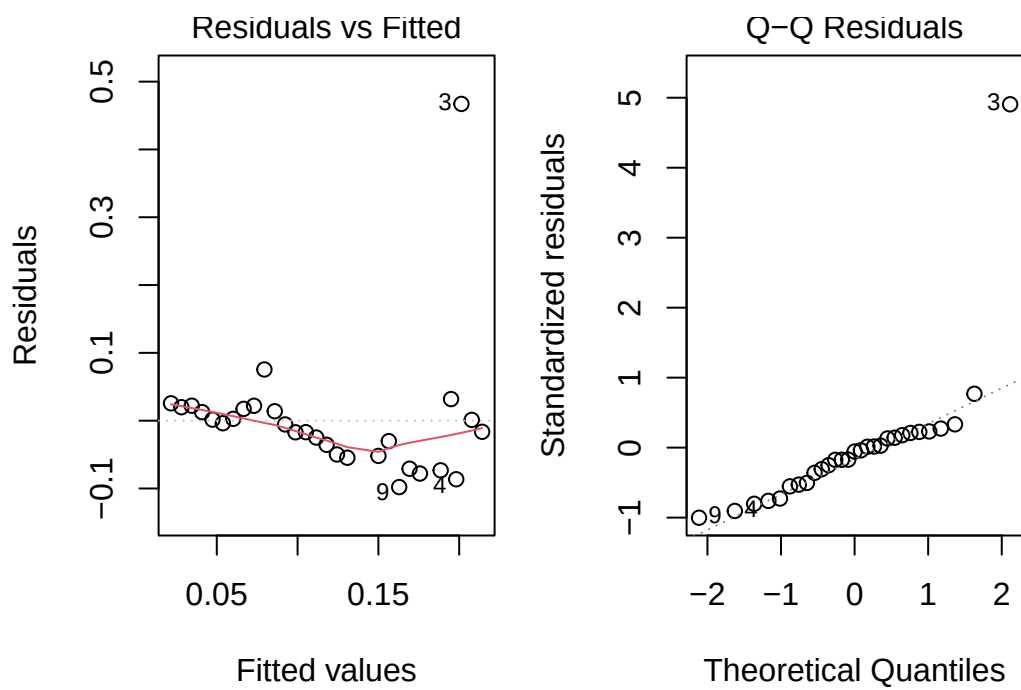
The analysis of Summer Olympic Games data from 1896 to 2016 reveals clear and statistically significant changes in participation and medal outcomes over time. Overall athlete participation has increased substantially, reflecting the long-term expansion of the Olympic Games and its growing global reach. Female success has also increased steadily, showing an upward trends over time. These results provide consistent evidence that the Olympics have become more inclusive, particularly in terms of gender representation.

In addition, the distribution of Olympic medals across nations has become less concentrated over time. The declining Herfindahl–Hirschman Index suggests that medals are no longer dominated by a small group of countries to the same extent as in earlier Olympic Games. This



(a) Residuals vs Fitted

Figure 4: Residual plots for female medal success model



(a) Residuals vs Fitted

Figure 5: Residual plots for HHI model

indicates increasing competitiveness and a broader international spread of sporting success, although traditional sporting powers still remain highly successful.

Overall, the findings highlight a clear long-term transformation of the Olympic Games: greater participation, improved gender balance, and more distributed international success. Together, these patterns reflect both the institutional growth of the Olympics and broader global changes in access to sport and investment in athletic development.

8 Limitations

This study has several limitations. The analysis focuses on long-term trends using Year as the primary explanatory variable and does not account for other factors that may influence participation and medal outcomes, such as population size, economic development, government investment in sport, host-country effects, or geopolitical events. In addition, the linear and logistic regression models may not fully capture non-linear changes in Olympic history, such as major expansions in events or disruptions like world wars. Finally, the HHI measure summarises medal concentration but does not explain the underlying reasons for changes in competitiveness across countries. Future research could explore more complex models that incorporate additional predictors and non-linear effects, as well as qualitative analyses to understand the social and political factors driving changes in Olympic participation and success.

References

Griffin, Randi. 2018. *120 Years of Olympic History: Athletes and Results*. <https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>.